Aaron Su

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Summary of Qualifications

- Electrical & Computer Engineering student specializing in machine learning for sensor and embedded systems
- Build across hardware and machine learning: multimodal deep learning in PyTorch, network protocols implemented from scratch, and PCB-level circuit design.
- Act as a connector between faculty, staff, and incoming researchers, building shared documentation and tooling that cut duplicated effort across a 120-person lab.
- Take projects from problem identification through deployment: spotted an unmanaged software licensing cost, built the monitoring for it, and shipped the decision tool that resolved it.
- Licensed Emergency Medical Responder; four years making decisions under real-time pressure.
- **Skills:** Python, Java, JavaScript, R, MATLAB | PyTorch, NumPy | Playwright, Git, Obsidian | KiCAD, MultiSIM, Fusion 360, PCB Design
- **Languages:** English (Native), Cantonese (Native), Mandarin (Limited working proficiency), Japanese (Elementary)

Education

UNIVERSITY OF WASHINGTON | SEATTLE, WA **SEPT 2024 - JUN 2028**

- *Bachelor of Science, Electrical & Computer Engineering*
- *Minor in Data Science, Minor in Informatics*
- Related Coursework: Computer Programming III, 3D Printing, Data Science I, Signals & Systems, Probability I, Cybersecurity, Computer-Communication Networks

Professional Experience

MOBILE, INTELLIGENT, AND NETWORKED SYSTEMS LAB, HKUST | HONG KONG, SAR **JUNE 2026 - PRESENT**

Research Intern, Machine Learning & Artificial Intelligence - Advised by Prof. Xiaomin Ouyang

- Building multimodal human activity recognition models in PyTorch for the CUHK-X Challenge (CUHK AIoT Lab, $10K prize pool, UbiComp 2026 finals); currently **top 30%** of an international leaderboard on a 40-class recognition task.
- Built preprocessing and encoder pipelines across three heterogeneous sensor streams (vision-derived 3D skeleton keypoints, wearable IMU accelerometer and gyroscope series, thermal infrared arrays), handling differing sampling rates, temporal alignment, and per-channel normalization.
- Designed and trained three sensor-specific encoders in PyTorch (spatio-temporal skeleton transformer, IMU encoder, thermal transformer) on data from **24 subjects**, unified by a confidence-weighted late-fusion layer.

SENSORS, ENERGY, AND AUTOMATION LABORATORY (SEAL) | SEATTLE, WA **SEPT 2025 - JUNE 2026**

Student Researcher, AI Engineer - Advised by Prof. Alexander V. Mamishev

- Built a Playwright automation that pulls software license usage data across shared workstations serving **60+ researchers**, replacing manual tracking with continuous monitoring.
- Designed a cost model on that usage data that flags when buying an additional license beats paying per-use costs, **saving the lab ~$1,500**.
- Deployed three prompt-engineered LLM assistants over a structured Obsidian knowledge base, cutting new-member onboarding time and repetitive support questions.
- Automated recurring lab operations in JavaScript and Google Apps Script across document handling, tracking, and reporting.

LICENSED EMERGENCY MEDICAL RESPONDER | CITY OF RICHMOND | RICHMOND, BC SEPT 2022 - PRESENT

- Treated 150+ patients and supervised response teams at events with 1,000+ attendees.

Projects

WI-FI PHYSICAL LAYER RECEIVER | Python, NumPy | aaronsuu.github.io/projects/wifi-transmitter

- Implemented an 802.11 receiver from scratch: OFDM demodulation via FFT, soft-decision Viterbi decoding over a rate-1/2 trellis, and matched-filter packet detection that stays reliable at negative SNR.

RELIABLE TRANSPORT PROTOCOL OVER UDP | Python, Sockets | aaronsuu.github.io/projects/reliable-udp-transfer

- Built a TCP-style transport layer with three-way handshake, sliding-window ARQ, per-packet ACKs, and file-ID multiplexing, sustaining 5+ concurrent transfers under 5% packet loss.